Lucas Witthauer

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Electrical Engineering student with hands-on experience in custom PCB design, embedded firmware, and power electronics gained through industry internship and coursework. Strong foundation in circuit analysis, mathematics, and computer engineering. Proven ability to learn quickly, communicate clearly, and balance demanding technical and athletic commitments in a fast-paced, security-conscious engineering environment.

TECHNICAL SKILLS

Programming Languages: C++, MicroPython, Python, Java, JavaScript, Assembly

Software & Tools: Altium Designer, LTspice, Arduino IDE, Wokwi, Microsoft Suite, Git/GitHub, Raspberry Pi

Core Competencies: PCB Design, Power Electronics, ITAR-Controlled Environments, Circuit Analysis, Embedded Systems, Digital Logic Design, Debugging, Bench Testing, Technical Communication

ENGINEERING & TECHNICAL EXPERIENCE

Electrical Engineering Intern | PowerLight Technologies – Kent, WA *Summer 2026*

- Worked within an ITAR-controlled engineering environment, handling export-controlled hardware and technical data in accordance with federal compliance requirements
- Designed and laid out a custom multi-zone resistive heater PCB in Altium Designer, including schematic capture, component footprint sourcing, and via treatment for a high-power power-electronics subsystem
- Selected and validated power components — MOSFETs, thermal isolation pads, and high-current terminal blocks — for a multi-zone resistive heating and PWM switching architecture
- Developed embedded firmware in C++ for a Teensy 4.1 microcontroller implementing multi-zone PWM control logic and closed-loop current sensing
- Built a companion Arduino/OLED visualization tool in Wokwi to model and demonstrate multi-zone power distribution profiles
- Performed hands-on bench testing and validation of PCB assemblies using lab power supplies and test equipment

IR Frequency Detector – Circuit Design | Seattle Pacific University *March 2026*

- Designed a filter for a 10 kHz IR beam using 3 cascaded Bandpass Sallen-Key stages with voltage dividers on each stage
- Connected the output of the 3rd filter stage to a peak detector circuit, fed into an LM339 comparator to control detection logic

Multi-Output Benchtop Power Supply – PCB, Altium Designer | Seattle Pacific University *Spring 2026*

- Designed and prototyped a dual adjustable power supply PCB delivering seven output rails using LM317/LM337 adjustable regulators and UA7812/UA7912/UA7805 fixed regulators
- Completed full project cycle: LTspice simulation, breadboard validation, PCB fabrication, hardware testing, and formal technical documentation

EDUCATION

B.S. in Electrical Engineering | Seattle Pacific University – Seattle, WA *Expected June 2027*

GPA: 3.36 | *Recipient, Andy Grove Scholarship for Intel Employees' Children* | *Computer Engineering Minor*

Relevant Coursework: Calculus I–III, Linear Algebra, Differential Equations, Physics I–III, Circuits I–II, Electronics, Electromagnetism, Computer Organization & Assembly, Digital Logic Design, Microcontrollers, Object-Oriented Programming, Data Structures

ADDITIONAL EXPERIENCE

NCAA Division II Student-Athlete – Men's Soccer | Seattle Pacific University *August 2023 – Present*

- Compete as a varsity student-athlete in the Great Northwest Athletic Conference, balancing 20+ hours/week of training, film review, travel, and competition with a full-time EE course load
- Maintained a 3.36 GPA while competing at the Division II level, reflecting consistent time management, discipline, and the ability to perform under pressure on tight deadlines
- Collaborate daily within a 30-person team environment, building communication, coachability, and reliability under high-stakes, high-visibility conditions
- Organized and led team logistics events, including coordinating a team breakfast for 35+ players and coaching staff, developing planning and cross-functional coordination skills
- Bring a team-first mindset, strong work ethic, and resilience from athletics into engineering coursework and internship work